

Balázs Szekér

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Education

MSc	ETH Zürich, Statistics - Master's thesis at the Learning & Adaptive Systems Group, supervised by Frederike Lübeck and Andreas Krause, on continual learning in multi-turn, tool-use environments. - Relevant coursework: - Machine learning & AI: Advanced Machine Learning, Probabilistic Artificial Intelligence, Foundations of Reinforcement Learning, Large Language Models, Big Data - Statistics: Fundamentals of Mathematical Statistics, Time Series Analysis, Computational Statistics	Zürich, Switzerland 2024 – present
BSc	ETH Zürich, Physics Bachelor's thesis at the Paul Scherrer Institute (PSI) in radiochemistry; the research contributed to a peer-reviewed publication.	Zürich, Switzerland 2019 – 2022

Experience

United Nations Office for Disaster Risk Reduction (UNDRR), AI Engineering Intern (LLMs & Retrieval Systems) - Project manager leading the design and development of DATUM, a RAG-based LLM system for large-scale knowledge access. - Designed a Python ingestion and preprocessing pipeline, including document chunking strategies. - Experimented with retrieval configurations to improve relevance and factual grounding. - Implemented embedding and vector search (OpenAI API, PostgreSQL, Azure AI Search). - Built a chatbot prototype for internal evaluation and deployment.	Bangkok, Thailand Oct 2025 – Mar 2026 6 months
Sika Technology AG, Research and Development, Data Scientist (Spatiotemporal Modeling) - Designed and implemented an application to forecast global roof-colour evolution from satellite imagery using spatiotemporal regression. - Developed a Python pipeline using the Google Earth API to retrieve high-resolution satellite images. - Adapted a physics-based correction model to remove atmospheric and surface-impurity artefacts from raw satellite data. - Built a time-series regression framework and used Gaussian Process Regression to model its hyperparameters spatially.	Zürich, Switzerland Mar 2025 – July 2025 5 months
ETH Zürich, Teaching Assistant (Machine Learning) - Probabilistic Artificial Intelligence (A. Krause): creating and revising the coding assignments, leading exercise sessions. - Introduction to Machine Learning (A. Krause & F. Yang): leading exercise sessions, developing and reviewing homework.	Zürich, Switzerland Feb 2025 – present 1 year 9 months
Empa, Swiss Federal Laboratories for Materials Science and Technology, Research Intern (Computer Vision) Developed a Python pipeline for vehicle detection and tracking in drone footage (optical flow), producing velocity and acceleration profiles for downstream analysis.	Dübendorf, Switzerland Oct 2022 – Dec 2022 3 months

Publications

Unraveling the formation of Tb/Pd films by coupled reduction as targets for heavy ion-beam irradiation

Oct 2024

N. Cerboni, B. Szekér, R. Dressler, et al.

[10.1016/j.jallcom.2024.176954](https://doi.org/10.1016/j.jallcom.2024.176954) (Journal of Alloys and Compounds, vol. 1010)

Projects

Brain age prediction from MRI features

Regression model estimating brain age from MRI-derived features. Ranked 13th of 150 teams.

Heart-rhythm classification from ECG data

CNN-based classifier for heart-rhythm anomalies in ECG signals. Ranked 14th of 131 participants.